A FINITE TYPE MODULE FLAT ALONG A FIBRE BECOMES FINITELY PRESENTED AND FLAT ON AN OPEN AFTER AN ELEMENTARY ETALE NEIGHBOURHOOD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The local structure of a finite type flat sheaf at a point. Same chapter flat.tex, label proposition-finite-type-flat-at-point, invoked at proof line 2767. Produces, for each point of the fibre, an elementary etale neighbourhood and an open subset of the base change containing the point on which the pullback is of finite presentation and flat.
- Existence of an elementary etale neighbourhood dominating finitely many given ones. More on Morphisms, label more-morphisms-lemma-elementary-etale-neighbourhoods, invoked at proof line 2781. Supplies one elementary etale neighbourhood of the base dominating the finitely many pointwise ones.

Source reference. Original lemma 057B: sources/stacks-chow/tags.txt, source lines 6739–6739. Navigation pointer for \ref{more-morphisms-lemma-elementary-etale-neighbourhoods} used inside the counted span (line 2781); the argument that consumes it is retained in the span.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 2767); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $s \in S$. Assume that*

(1) *f is of finite presentation,*

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(2) $\mathcal{F}$ *is of finite type, and*
(3) $\mathcal{F}$ *is flat over* S *at every point of the fibre* X_s.

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ *and an open subscheme*

$$V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$$

which contains the fibre $X_s = X \times_S s'$ *such that the pullback of* $\mathcal{F}$ *to* V *is an* $\mathcal{O}_V$*-module of finite presentation and flat over* $\mathcal{O}_{S',s'}$.

Proof. For every point $x \in X_s$ we can use Proposition 05I5 to find an elementary étale neighbourhood $(S_x, s_x) \to (S, s)$ and an open $V_x \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S_x,s_x})$ such that $x \in X_s = X \times_S s_x$ is contained in V_x and such that the pullback of $\mathcal{F}$ to V_x is an $\mathcal{O}_{V_x}$-module of finite presentation and flat over $\mathcal{O}_{S_x,s_x}$. In particular we may view the fibre $(V_x)_{s_x}$ as an open neighbourhood of x in X_s. Because X_s is quasi-compact we can find a finite number of points $x_1, \ldots, x_n \in X_s$ such that X_s is the union of the $(V_{x_i})_{s_{x_i}}$. Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ which dominates each of the neighbourhoods (S_{x_i}, s_{x_i}), see More on Morphisms, Lemma 057B. Set $V = \bigcup V_i$ where V_i is the inverse images of the open V_{x_i} via the morphism

$$X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'}) \longrightarrow X \times_S \mathrm{Spec}(\mathcal{O}_{S_{x_i},s_{x_i}})$$

By construction V contains X_s and by construction the pullback of $\mathcal{F}$ to V is an $\mathcal{O}_V$-module of finite presentation and flat over $\mathcal{O}_{S',s'}$. □